LITERATURE REVIEW

Literature Review

1. Feedback & Evaluation

The EchoSign project will be evaluated based on its ability to recognize Arabic sign language gestures and convert them into readable text. The evaluation will consider the project's objectives, technical implementation, user interface, and overall usability.

Lecturer feedback will be collected during project reviews and presentations. The feedback will help the team identify strengths, address technical issues, and improve the quality of the final system.

The evaluation will focus on:

- The relevance of the project to the communication needs of Arabic sign language users.
- The accuracy of recognizing dynamic signs representing individual words.
- The effectiveness of the machine learning model and its integration with the online server or API.
- The usability, accessibility, and reliability of the Flutter mobile application.
- The quality of project documentation, testing, and presentation.

2. Suggested Improvements

Several improvements can be considered to enhance the performance and usability of EchoSign:

- **Improve Recognition Accuracy:** Expand and diversify the dataset to include different signers, lighting conditions, backgrounds, and signing speeds.
- **Support More Signs:** Increase the number of Arabic sign language words supported by the recognition model.
- **Optimize Model Performance:** Reduce prediction latency and improve inference efficiency to provide faster translations.
- **Enhance API Reliability:** Improve communication between the mobile application and the online server and handle network errors appropriately.
- **Improve User Experience:** Provide clear feedback during sign recognition and display understandable error messages when a sign cannot be recognized.
- **Expand Testing:** Test the application with different users and devices to identify usability issues and evaluate real-world performance.
- **Future Development:** Consider supporting complete sentences, continuous sign recognition, and additional Arabic sign language dialects in future versions.

3. Final Grading Criteria

The following is a proposed grading distribution for evaluating the EchoSign project. The final criteria and percentages should be adjusted according to the lecturer's official assessment rubric.

Assessment Component	Description	Proposed Marks
Documentation	Project proposal, literature review, requirements, design, and technical documentation	20%
Implementation	Mobile application, machine learning model integration, and API communication	35%
Testing & Evaluation	Functional testing, model accuracy evaluation, and performance testing	20%
Presentation& Demonstration	Presentation quality, live demonstration, and explanation of technical decisions	15%
Teamwork & Project Management	Task distribution, progress tracking, and adherence to the project plan	10%
Total		100%